
Introduction to Machine Learning

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Notice

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Outline

- Definitions and examples
- What are the types of learning problems?
 - Supervised learning
 - Unsupervised learning
 - Reinforcement learning
- Resource recommendations
- Review Syllabus
- Discuss who should take this course

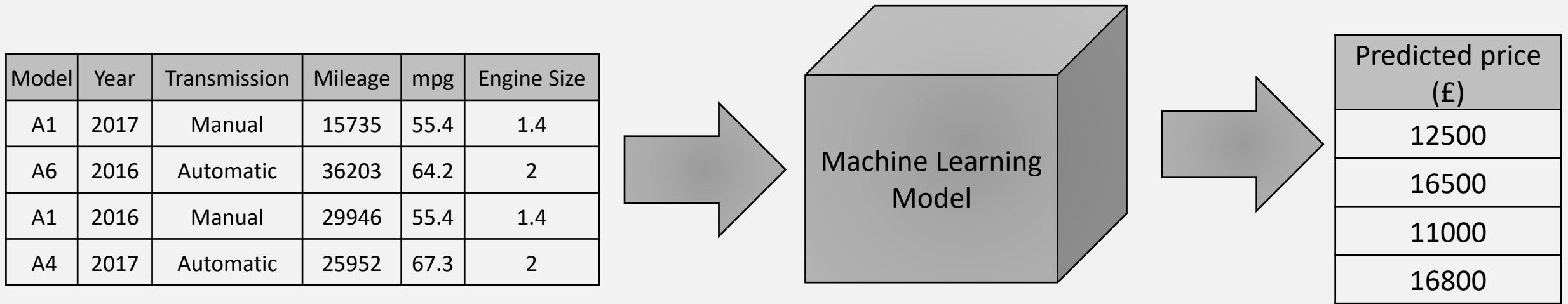
Definitions and Notations

Related Terms and Fields

- Artificial Intelligence
- Statistical learning and probability theory
- Pattern recognition
- Signal processing
- Data mining
- Information theory
- Optimization and linear algebra
- Business analytics

Price Prediction Example – Car Prices

- Task: given a car's description, predict its average market value.



Input is a list of car descriptors.
Each row represents a car.

See: www.kaggle.com/adityadesai13/used-car-dataset-ford-and-mercedes

What is Machine Learning? [1]

- From previous example, there is a specific **task**.
- The task has an **input** and an **output**.
- There is a desired goal for the task; e.g. predict an event with high **performance**.
- The machine learning model performs the task to achieve the desired goal with high performance.
- Performance measure is task dependent (more on this in later lectures).

What is Machine Learning? [2]

- “Machine Learning is the study of algorithms that improve their **performance P** at some **task T** with **experience E**. A well-defined learning task is given by $\langle P, T, E \rangle$.” – Tom Mitchell. Machine Learning (Book). 1997.
- “Vast amounts of data are being generated in many fields, and the statistician’s job is to make sense of it all: to extract important patterns and trends, and understand “what the data says.” We call this learning from data.” – Elements of statistical learning (Book). 2012.
- Machine learning is the use of data to build a program that performs a specific task. The more data we have, the better the program gets at the task.

Learning from data – Car price prediction [1]

- **Task:** given a car's description, predict its average market value.
- **Performance** measure: **accuracy** of predicted value.
- **Experience:** Historical car price data. Suppose we have 1000 prices from past 6 months. Every row corresponds to an **example/sample/instances**.

Model	Year	Price	Transmission	Mileage	FuelType	Tax	mpg	Engine Size
A1	2017	12500	Manual	15735	Petrol	150	55.4	1.4
A6	2016	16500	Automatic	36203	Diesel	20	64.2	2
• • •								
A3	2019	17300	Manual	1998	Petrol	145	49.6	1

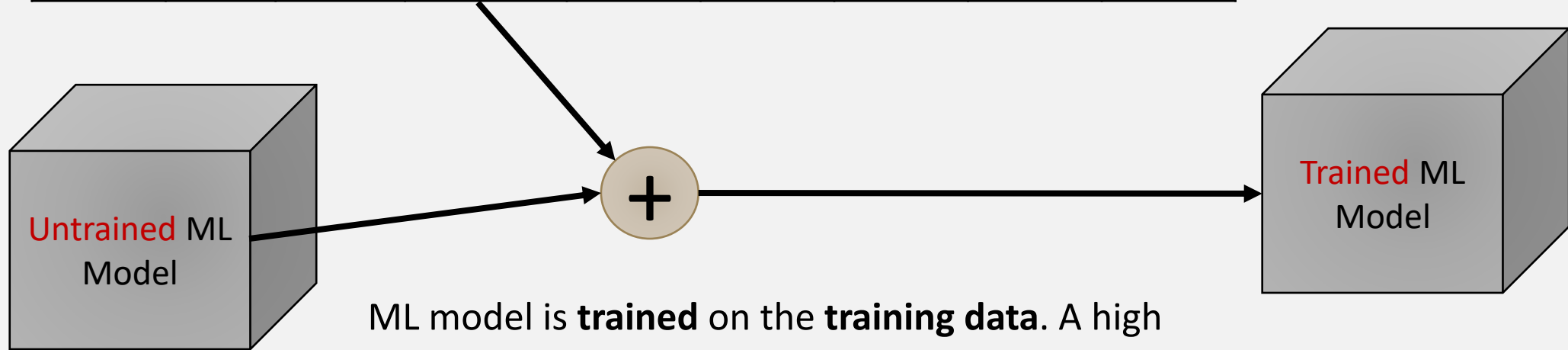
1000 **examples** to learn from.

Full dataset: www.kaggle.com/adityadesai13/used-car-dataset-ford-and-mercedes

Learning from data – Car price prediction [2]

Model	Year	Price	Transmission	Mileage	FuelType	Tax	mpg	Engine Size
A1	2017	12500	Manual	15735	Petrol	150	55.4	1.4
A6	2016	16500	Automatic	36203	Diesel	20	64.2	2
⋮								
A3	2019	17300	Manual	1998	Petrol	145	49.6	1

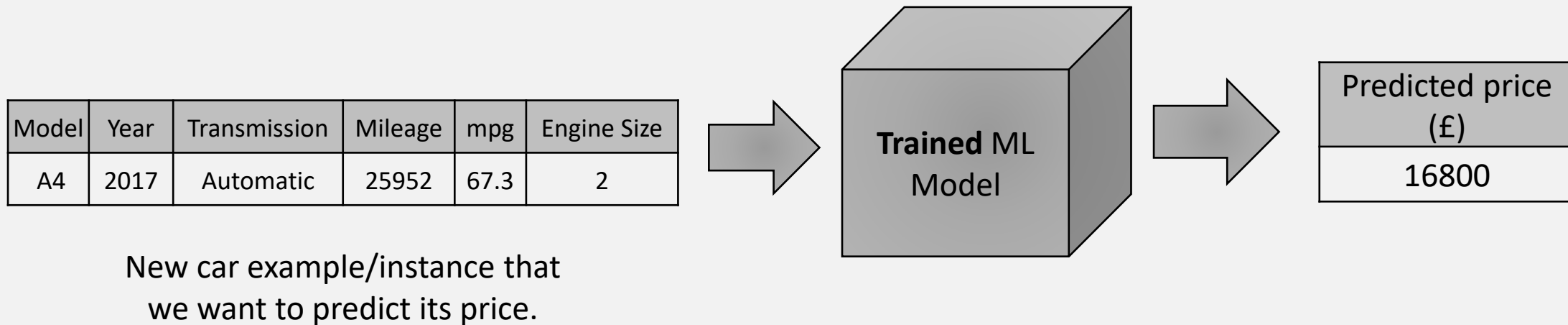
1000 **examples** to learn from.



ML model is **trained** on the **training data**. A high performing **Trained** model is produced.

Learning from data – Car price prediction [3]

- Trained ML model is used in production to predict on **new** instances. By new, we mean not part of the training data. So, the model has not been specifically trained on that instance.



What are features and labels?

- **Features**, input features, or input descriptors are the **inputs** to the machine learning model for a single example.
- **Labels** are the **output** of the ML model.
- Model, Year, Transmission, Mileage, mpg, and EngineSize are the **features** for the Car price prediction learning problem. The price is the **label**.
- Types of features (simplified): strings and numbers.

string	number	string	number	number	number
Model	Year	Transmission	Mileage	mpg	Engine Size
A4	2017	Automatic	25952	67.3	2

number
Predicted price (£)
16831

- All features and labels need to be represented in a number format to be processed by the model (computer program). More on this later.

Math notation for instances/examples

- On paper, we use mathematical notation for expressing instances and the learning problem.
- Common notation, we represent each instance as a fixed-length **vector** using the letter x .
- The labels are represented as a single-valued vector (scalar) using the letter y .

Model	Year	Transmission	Mileage	mpg	Engine Size
A4	2017	Automatic	25952	67.3	2

True price (£)
16800

- Thus, this instance is expressed as:

$x = [A4, 2017, Automatic, 25952, 67.3, 2]$ and $y = [16800]$

Math notation for a dataset

- A dataset is a set $\mathcal{S}$ with m instances (also called samples or examples).
- Each instance has a feature vector x and label y .
- $\mathcal{S} = \{ (x_1, y_1), (x_2, y_2), (x_3, y_3), \dots, (x_m, y_m) \}$

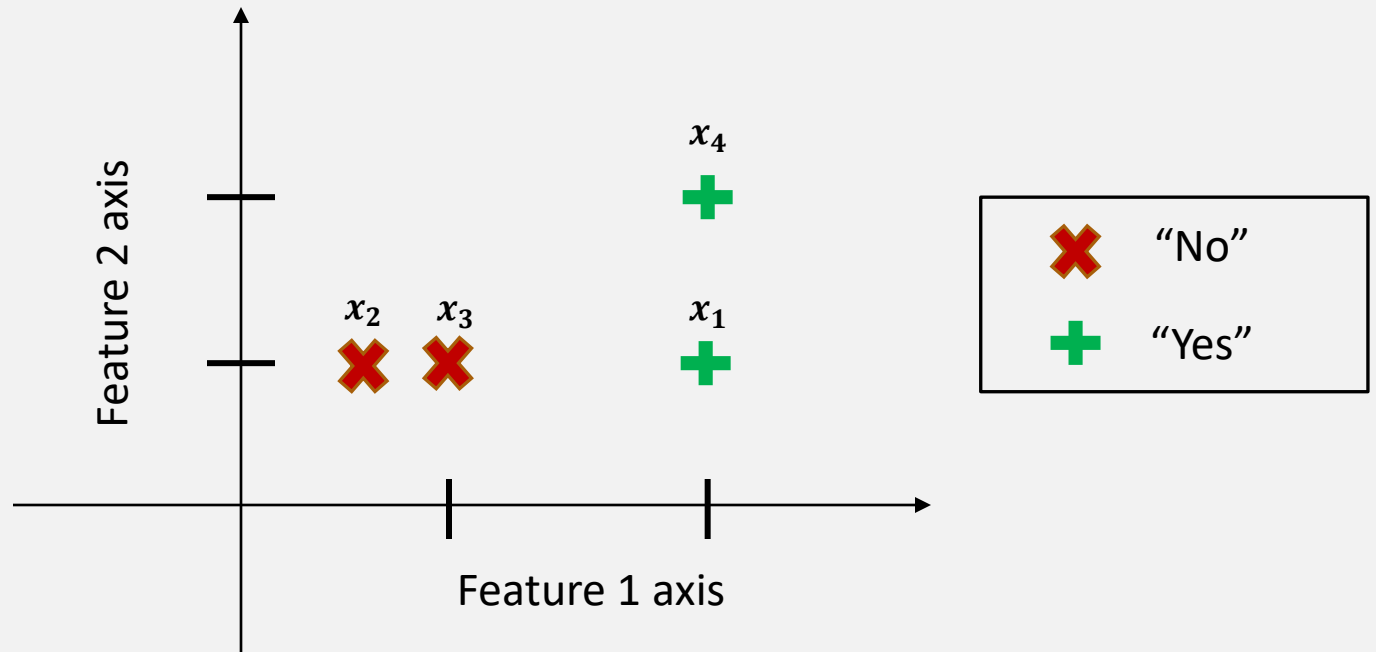
	Model	Year	Transmission	Mileage	FuelType	Tax	mpg	Engine Size		Price
x_1	A1	2017	Manual	15735	Petrol	150	55.4	1.4	y_1	12500
x_2	A6	2016	Automatic	36203	Diesel	20	64.2	2	y_2	16500
				.						.
				.						.
				.						.
x_m	A3	2019	Manual	1998	Petrol	145	49.6	1	y_m	17300

Represents a dataset. We can refer to this as a labeled dataset since the labels are known.

Feature space

- Using vector notation, the instances represent a point in n -dimensional cartesian space.
- For example, the below classification dataset has 2 features. Each instance represents a point in 2-dimensions.

	Feature 1	Feature 2	Label
x_1	2	1	Yes
x_2	0.5	1	No
x_3	1	1	No
x_4	2	2	Yes



Helpful Matrix Notation

- We can represent a dataset $\{(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots, (x_m, y_m)\}$ with m examples and n features as a feature matrix $X \in \mathbb{R}^{m \times n}$ and a label vector $Y \in \mathbb{R}^m$

$$X = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{bmatrix} = \begin{bmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_m^T \end{bmatrix} \quad \bigg| \quad Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}$$

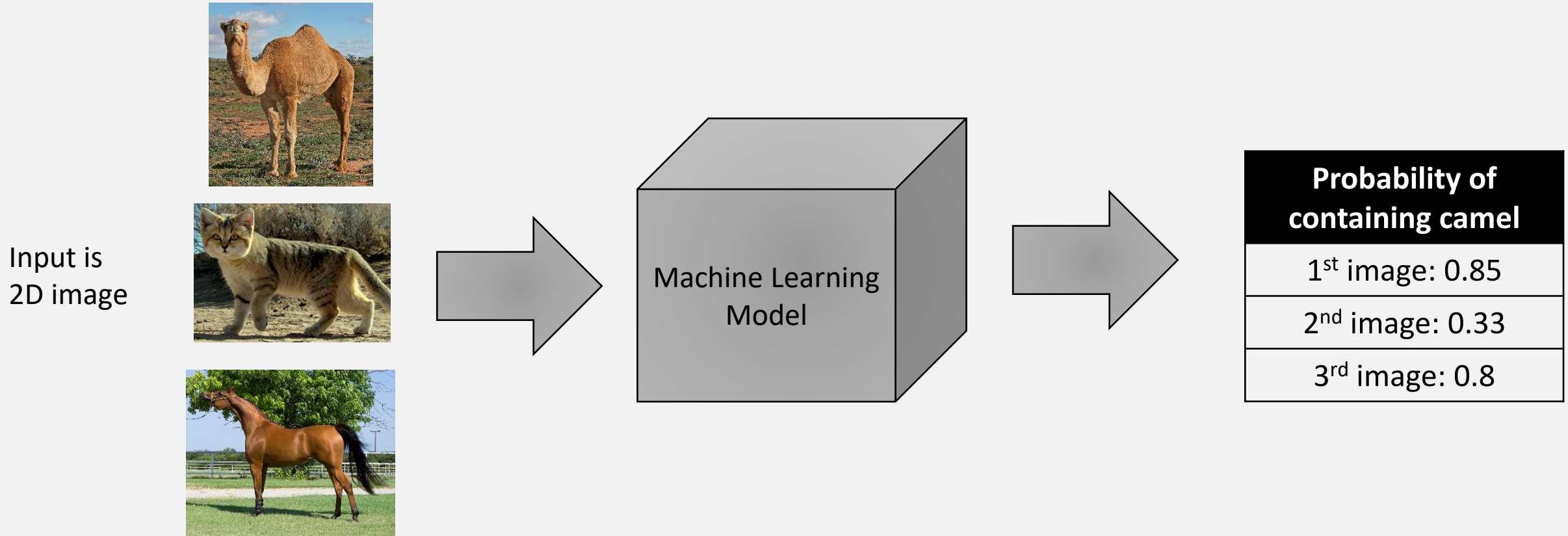
- X is referred to as the feature or data matrix. Y is the label or output vector.

Machine Learning – Problem Types

- **Supervised Learning:** learning problem involves the use of input features to predict a label. Goal is to train a model to predict labels from input features.
- **Unsupervised Learning:** Learning problem does not have labels, only features. The goal is to analyze the data and find interesting patterns.
- **Reinforcement Learning:**
 - Learning problem involves an agent that interacts with an environment. The agent can perform actions at every time step.
 - An agent's action results in a change in the environment.
 - The agent is rewarded or penalized based on the actions.
 - Through self-training in an environment and receipt of rewards, the agent gains experience and learns. With enough experience, the agent selects the best action given the situation.

SL Example – Image Recognition

- Task: given an image, predict if the image contains a camel.



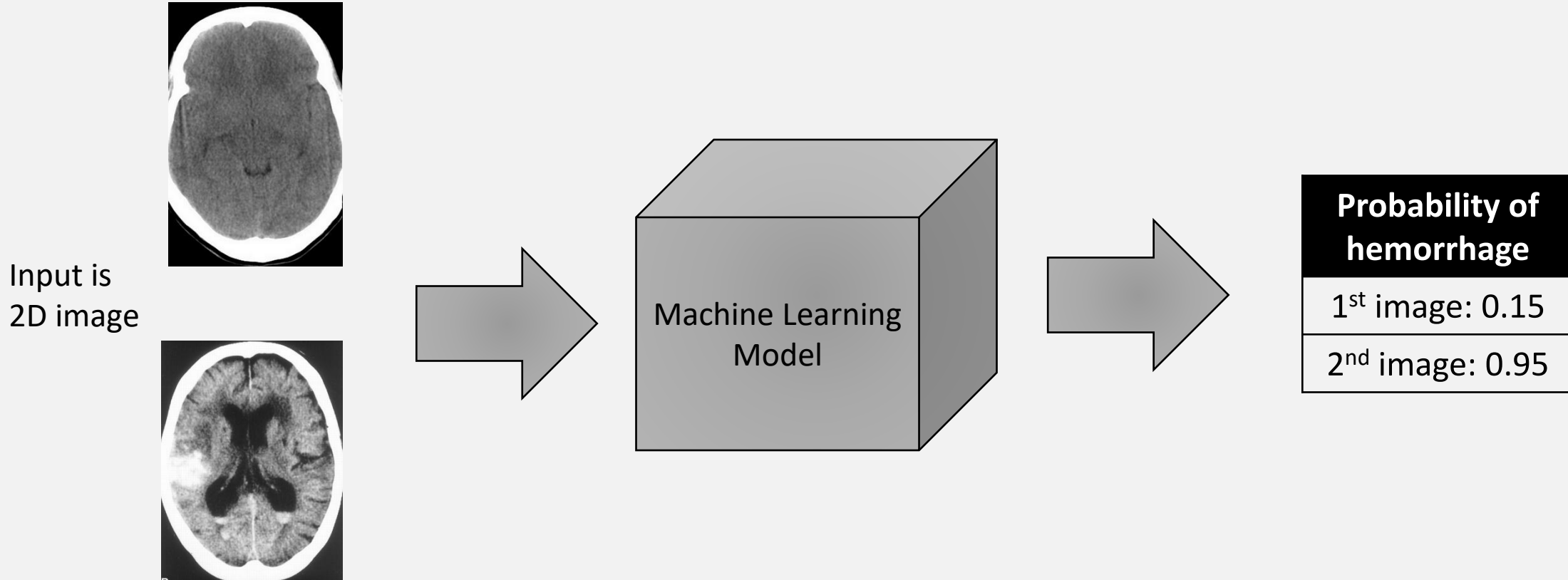
By Jjron - Own work, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=2831408>

By Payman sazesh - camera trap, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=24849743>

By Ealdgyth - Own work, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=4165040>

SL Example – Hemorrhage Diagnosis

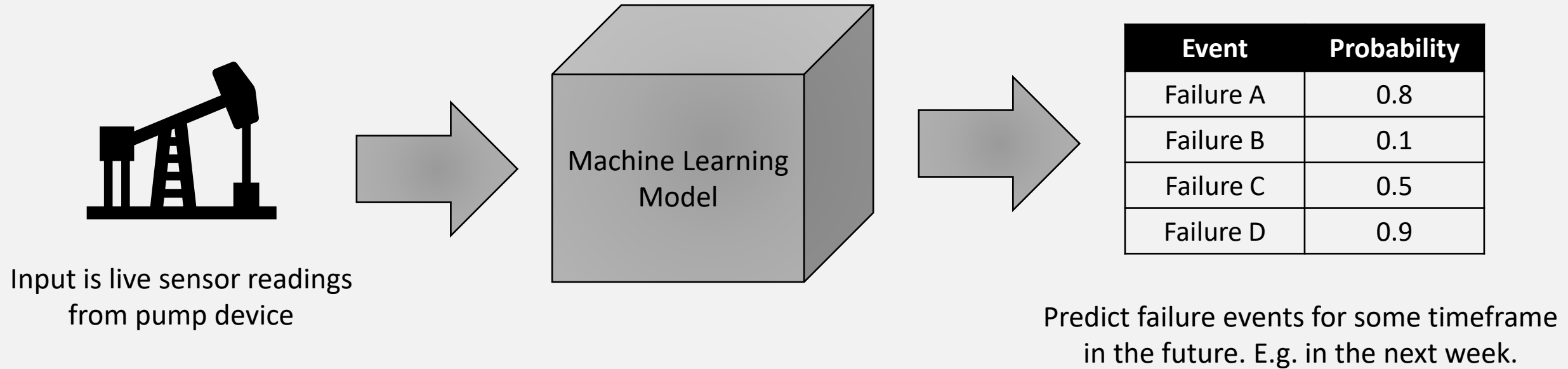
- Task: given a CT head scan, determine if there is hemorrhage.



See: www.kaggle.com/felipekitamura/head-ct-hemorrhage

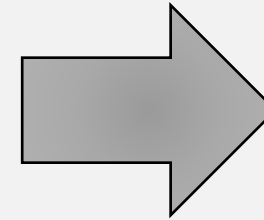
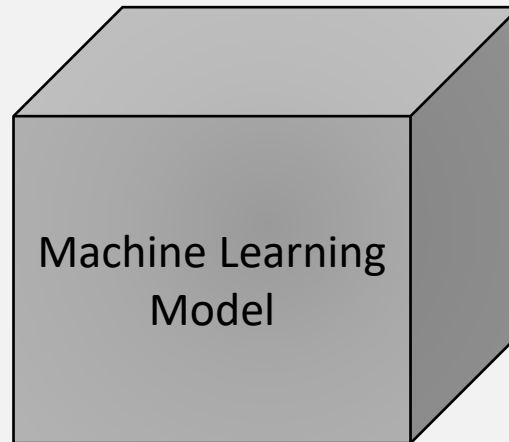
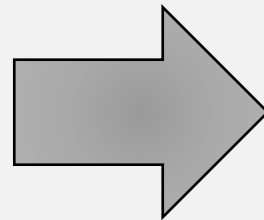
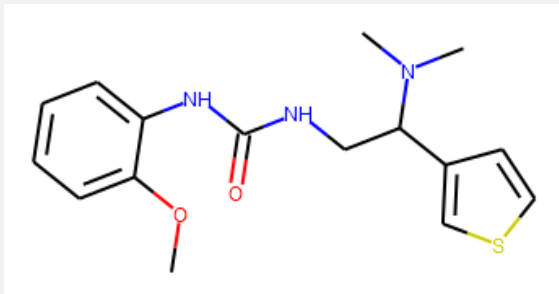
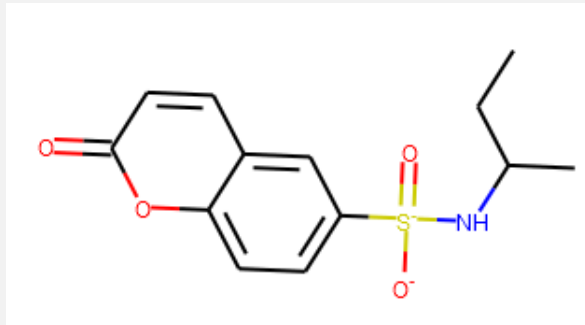
SL Example – Oil & Gas – Detecting early failure of machines

- Task: given live sensor readings of a device, predict if a failure will occur in the future.



SL Example – Drug Discovery Example – Finding new medicine

- Task: given a molecule, predict if it is effective against a biological target of interest. For example, we want to find molecules that work against a protein in the body that plays a role in a disease.



**Probability effecting
biological target**

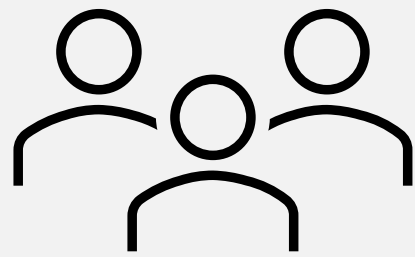
1st molecule: 0.3

2nd molecule: 0.99

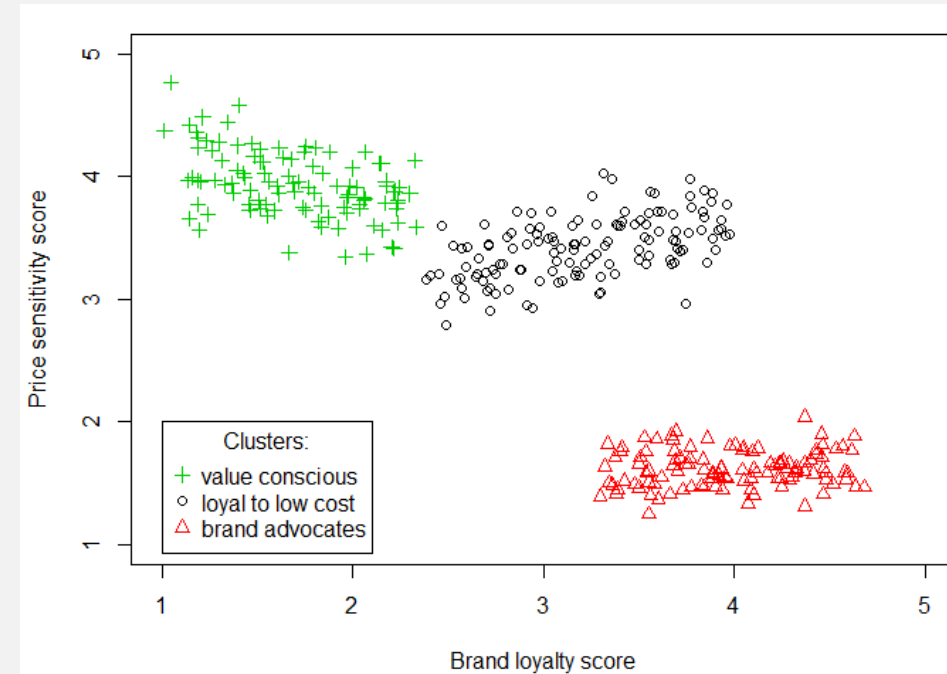
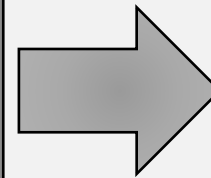
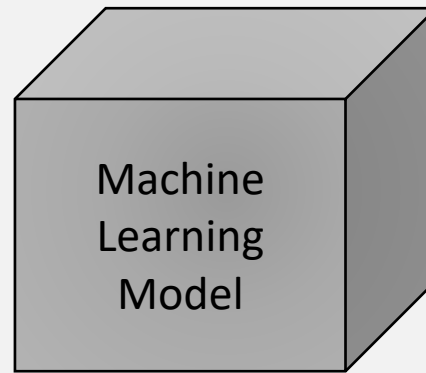
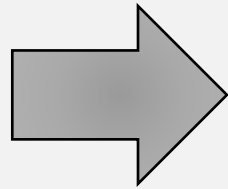
Input is computer representation
of a molecule; e.g. 2D image.

UL Example – Customer Segmentation/Clustering

- Task: given customer information, find interesting patterns to enhance business decisions. Notice there are no labels, just features.



Customer data



RL Example – Learning Strategy for Environment

- Task: given a complex environment, find the optimal strategy for the robot agent.

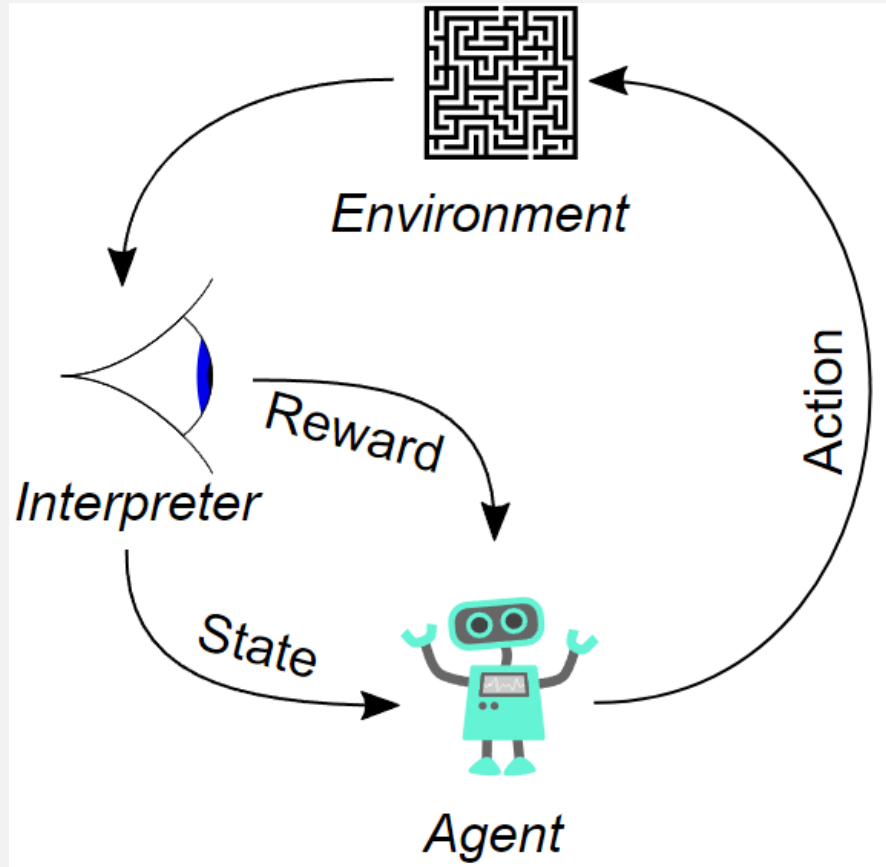
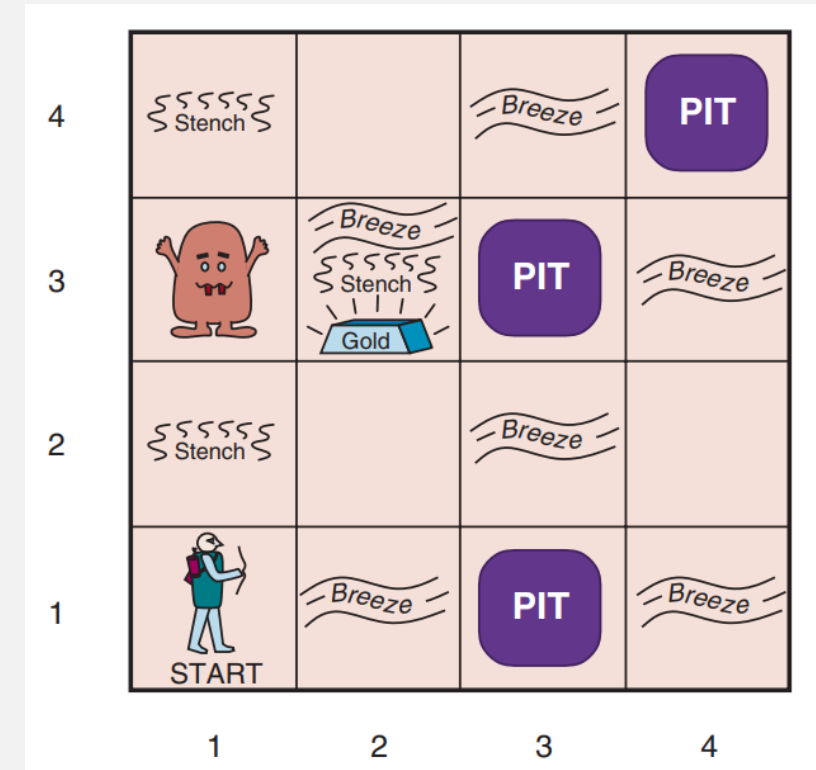


Image from [link](#)



Wumpus world toy problem

Supervised Learning

Simplified Introduction

Supervised Learning Overview [1]

The supervised learning problem consists of:

- **Labeled Dataset:** A dataset set $S = \{(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots, (x_m, y_m)\}$ consists of m instances (or examples).
- **Features:** Each x_i for $i = 1, 2, 3, \dots, m$ contains information about the instance. E.g. numbers, strings, etc.
- **Labels:** Each y_i for $i = 1, 2, 3, \dots, m$ is **assumed** to be a **function** of x_i . That is:

$$y_i = f(x_i)$$

- In practice, the dataset S is gathered from a real-world process.

Supervised Learning Overview [2]

The supervised learning problem consists of (cont.):

- **Machine Learning Model:** A machine learning model that produces a function h that is an **approximation** of the labeling function f .
- Question: Why doesn't the ML model just set $h = f$?
- Answer: The labeling function f is **unknown**.
- **Goal:** The ML model uses the labeled dataset S to produce $h \approx f$.

Supervised Learning with an example [1]

	Model	Year	Transmission	Mileage	FuelType	Tax	mpg	Engine Size		Price
x_1	A1	2017	Manual	15735	Petrol	150	55.4	1.4	y_1	12500
x_2	A6	2016	Automatic	36203	Diesel	20	64.2	2	y_2	16500
				•						•
				•						•
				•						•
x_m	A3	2019	Manual	1998	Petrol	145	49.6	1	y_m	17300

1. Suppose this labelled dataset $\mathcal{S}$ of $m = 1000$ instances was gathered from the real-world.

2. Suppose there is an **unknown** labelling function f such that:

$$y = f(\text{Model}, \text{Year}, \text{Transmission}, \text{Mileage}, \text{FuelType}, \text{Tax}, \text{mpg}, \text{EngineSize})$$

More specifically:

$$y_i = f(x_i) \text{ recall that } x_i \text{ is a vector. Note there are 8 column features for this specific dataset.}$$

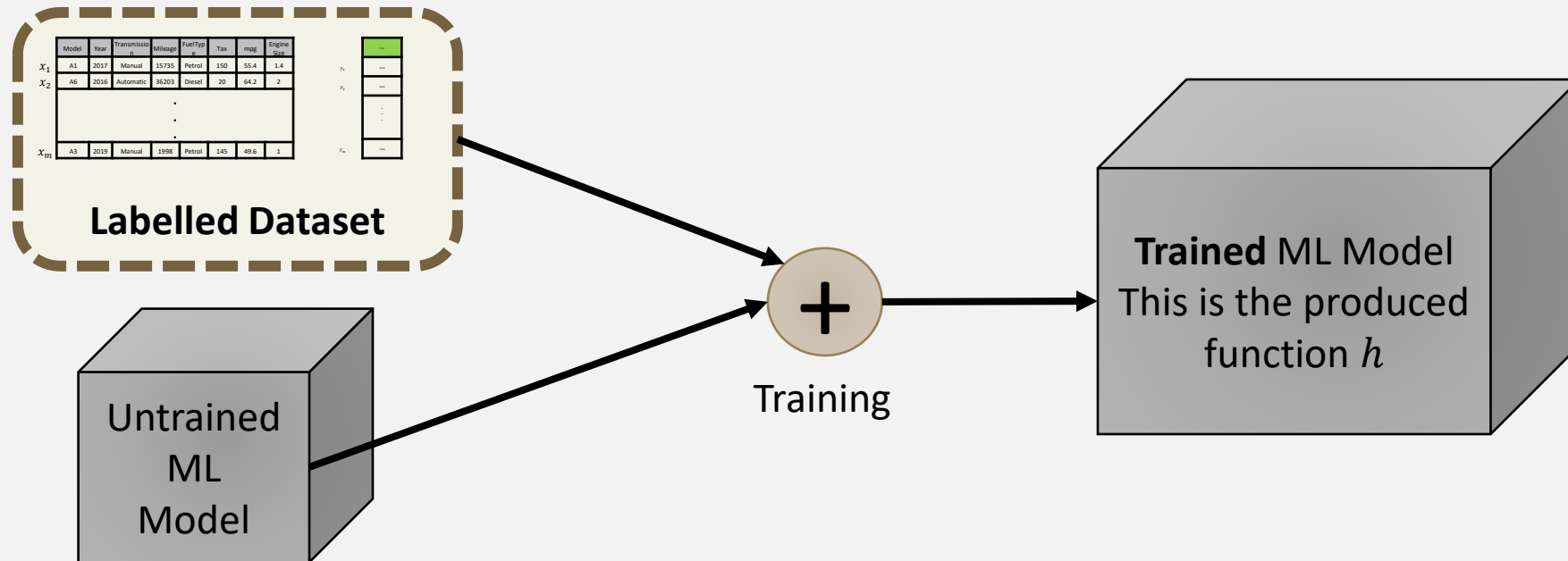
$$= f(x_{i1}, x_{i2}, x_{i3}, x_{i4}, x_{i5}, x_{i6}, x_{i7}, x_{i8}) \text{ where } x_{ij} \text{ denotes the } j\text{th feature value of the } i\text{th instance.}$$

Supervised Learning with an example [2]

So, we have this labelling function whose form or format is unknown:

$$y_i = f(x_{i1}, x_{i2}, x_{i3}, x_{i4}, x_{i5}, x_{i6}, x_{i7}, x_{i8})$$

3. The goal of the ML model is to use this labelled dataset to produce a function h that approximates f .



4. Now that we have h , then we can predict the label of new instances: $h(x_{new})$. Such that: $h(x_{new}) \approx f(x_{new})$

The Two Types of Supervised Learning

- **Regression:** The output labels are **continuous** real values. E.g. prices, scores, sensor readings, temperature, etc.
- **Classification:** The output label is a value from a small finite set.
 - **Binary classification:** output label is assigned a single value from a set of size 2. For example {No, Yes} or {0, 1}.
 - **Multiclass classification:** output label is assigned a single value from a set of size 3 or more. For example {small, medium, large} or {left, right, forward, down}.
 - **Multilabel classification:** output label is assigned multiple values from a set of size 2 or more. For example, when detecting multiple failure events for a single device; a device can suffer from multiple failures.

Supervised Learning Summary

- **Labelled dataset** gathered from real-world process.
- Machine learning model that **trains** on the labelled dataset to produce an approximator function h .
- New data without labels can now be passed to the trained ML model to compute predictions: $h(x_{new})$.
- Supervised learning types: regression and classification.
- We will revisit the supervised learning setup in a more formalized manner.

Unsupervised Learning Introduction

Unsupervised Learning Overview [1]

The **unsupervised** learning problem consists of:

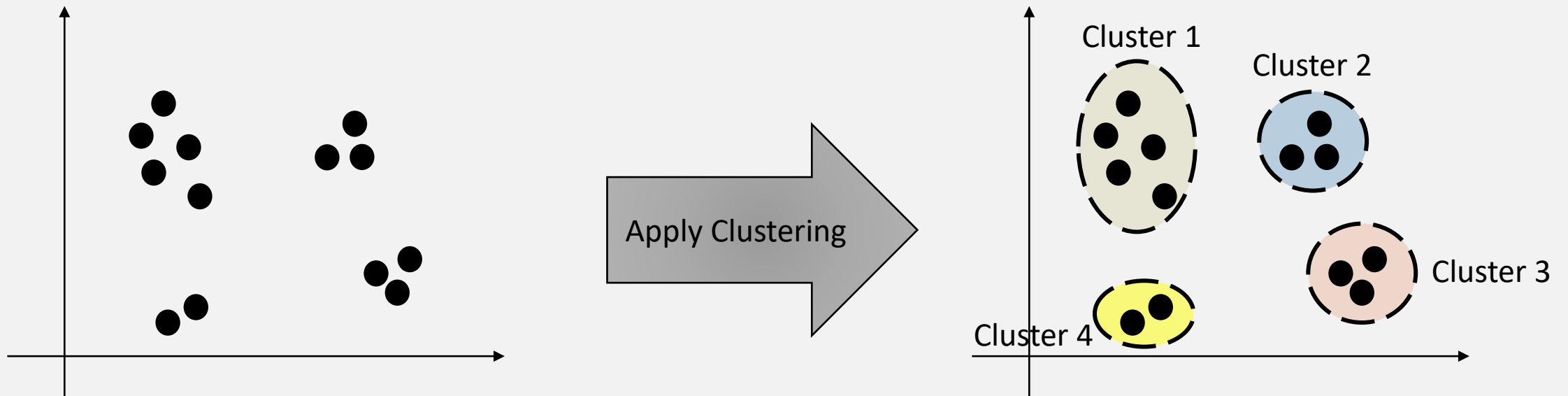
- **Unlabelled Dataset:** A dataset set $S = \{x_1, x_2, x_3, \dots, x_m\}$ consists of m instances (or examples).
- Note that we don't have any labels y .
- In practice, the dataset S is gathered from a real-world process.
- **Goal:** Find interesting patterns, structure, or groupings in the data.
- Common unsupervised tasks: clustering, anomaly detection, and dimensionality reduction.

Clustering (1)

- Given $\mathcal{S} = \{x_1, x_2, x_3, \dots, x_m\}$, group the instances into groups based on their distances to each other.
- How do we measure distance between x_i and x_j ? One simple way is via vector distance. Many other ways that is a function of the features.
- How many groups do we split the data into? Can fix this to a specific number, or make it adaptive based on a performance measure.

Clustering (2)

- Suppose we have a dataset with just 2 features and 13 instances. We want to cluster into 4 groups/clusters.

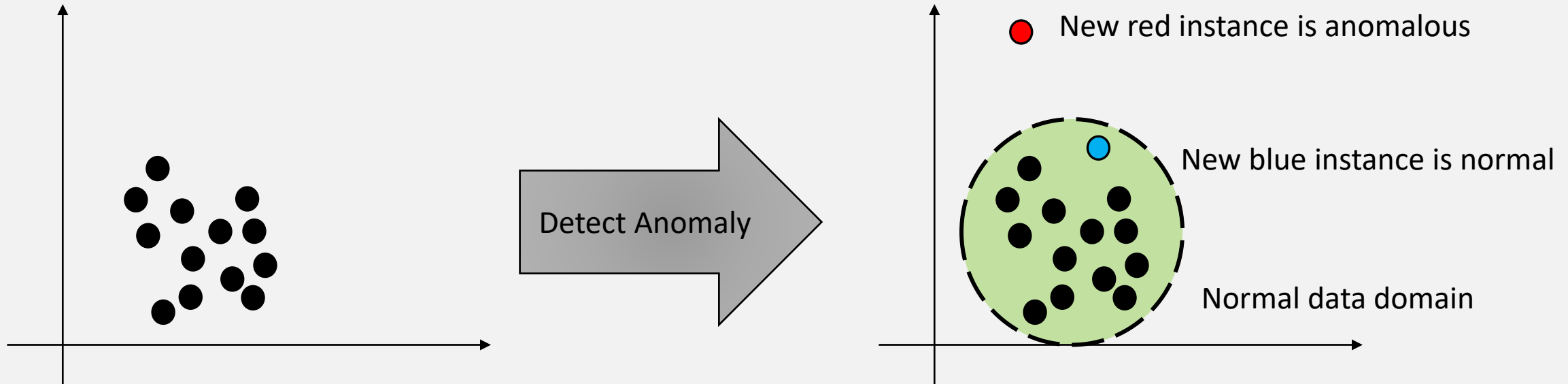


- Can analyze each cluster or group in more detail.
- Common application: visualization, consumer habits analysis, social network analysis.

Anomaly detection (1)

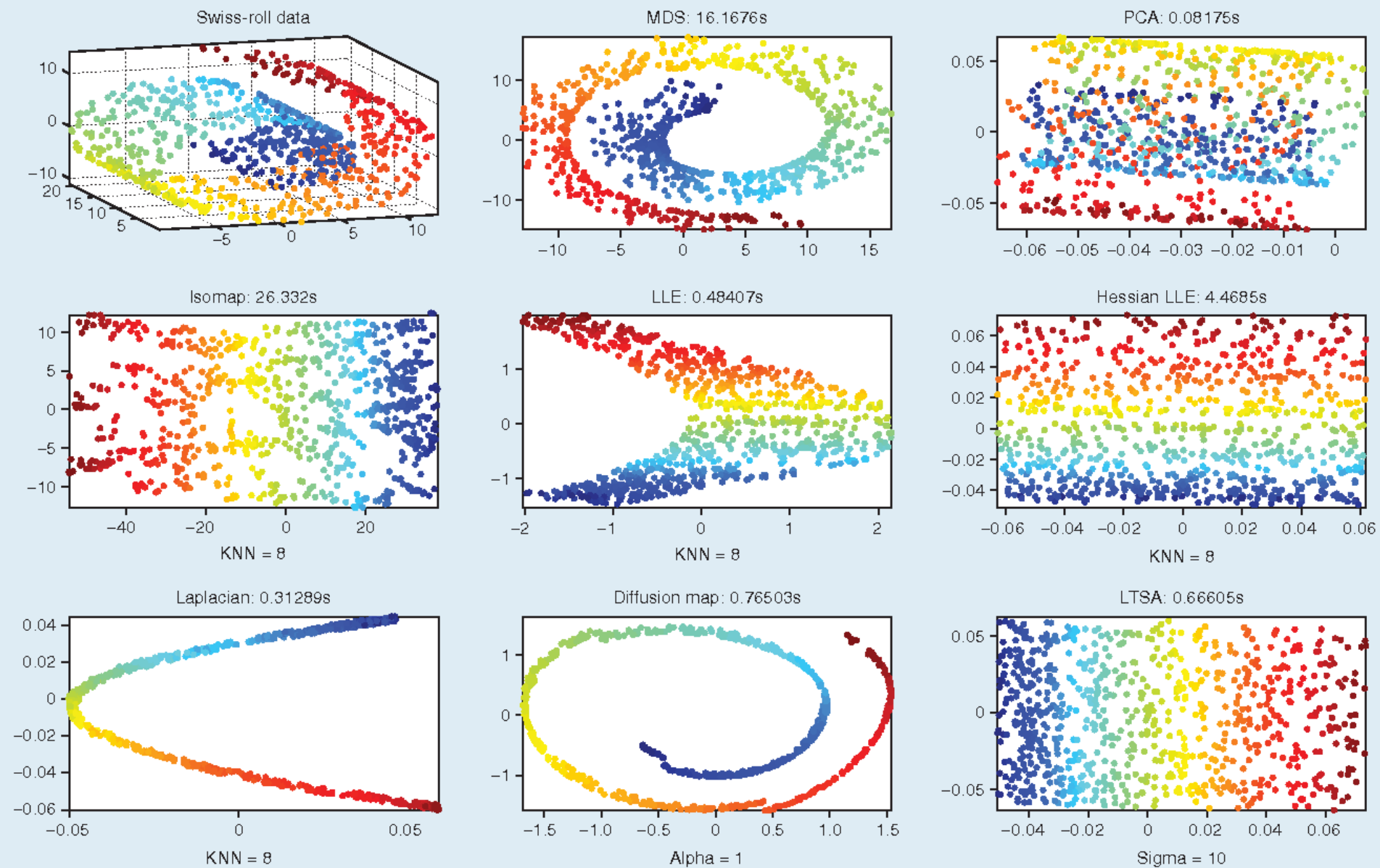
- Given $\mathcal{S} = \{x_1, x_2, x_3, \dots, x_m\}$, train a model to learn the **normal** pattern of an instance.
- The trained model can then be used to judge if a new instance is an outlier (anomaly, NOT normal, alarming).
- Can be used to detect alarming situations.
- Example 1: we have data on the purchasing habits of a person, we can detect if he purchases something that is out of the norm.
- Example 2: we have data from multiple sensors attached to oil pumps, we can detect if an oil pump's sensor readings are anomalous.
- Note that as we gather data over time, what is normal can shift. E.g. the purchasing habits of a person can change over time.

Anomaly Detection (2)



Dimensionality Reduction (1)

- Given $\mathcal{S} = \{x_1, x_2, x_3, \dots, x_m\}$, where x_i has d features, reduce the number of features from d to k .
- Typically, the data is mapped to lower dimensions while preserving the *structure* or *distance* of instances to each other.
- Common application is for visualization. That is map, the dataset from d dimensions to 2 or 3 dimensions and plot.
- The higher the dimensions, the harder it is to reduce to 2 or 3 dimensions while preserving the structure of the data.
- Other common application is to reduce the memory size of a high dimensional dataset. Fewer features -> smaller memory size.



3D Swiss-roll data (top-left) is reduced to 2D using eight dimensionality reduction techniques.

Image from: <https://ieeexplore.ieee.org/document/5401149>

Recommended Software

Programming Environment

- Python:
 - Easy to learn and use.
 - Fast prototyping. Can get code up and running fast. Little foundation required for new projects.
 - Lots of useful open-source libraries. Linear algebra, data management, plotting, machine learning, deep learning , etc.
- Recommend to install [anaconda](#) which is just a package manager for python:
 - Free version is enough for most users. Install anaconda, then follow the [tutorial](#) on setting up a python environment.
 - Manages installation of packages easily through the conda command.
 - Has IDE spyder environment. Or, you can use simple notepad++ for writing your code and run it using cmd. Or run interactive jupyter notebook.
 - Use the yml file to setup this course's environment; will release on blackboard.

ML python libraries

- numpy: linear algebra library for operating on vectors and matrices.
- pandas: library for manipulating datasets in easy to view format.
- scikit-learn: popular machine learning library with lots of ML algorithms.
- tensorflow: popular library for creating computational graphs. Many ML algorithms are computational graphs. Useful for other domains like optimization.
- keras: popular deep learning library. Recently incorporated into tensorflow.
- pytorch: popular deep learning library. Recently more popular than keras.
- matplotlib: popular plotting library for visualizing data or results.
- seaborn: another popular plotting library but with better visuals.
- gym: popular reinforcement learning library with lots of environments for testing your RL algorithms. Most likely not needed for the course.

scikit-learn algorithm cheat-sheet

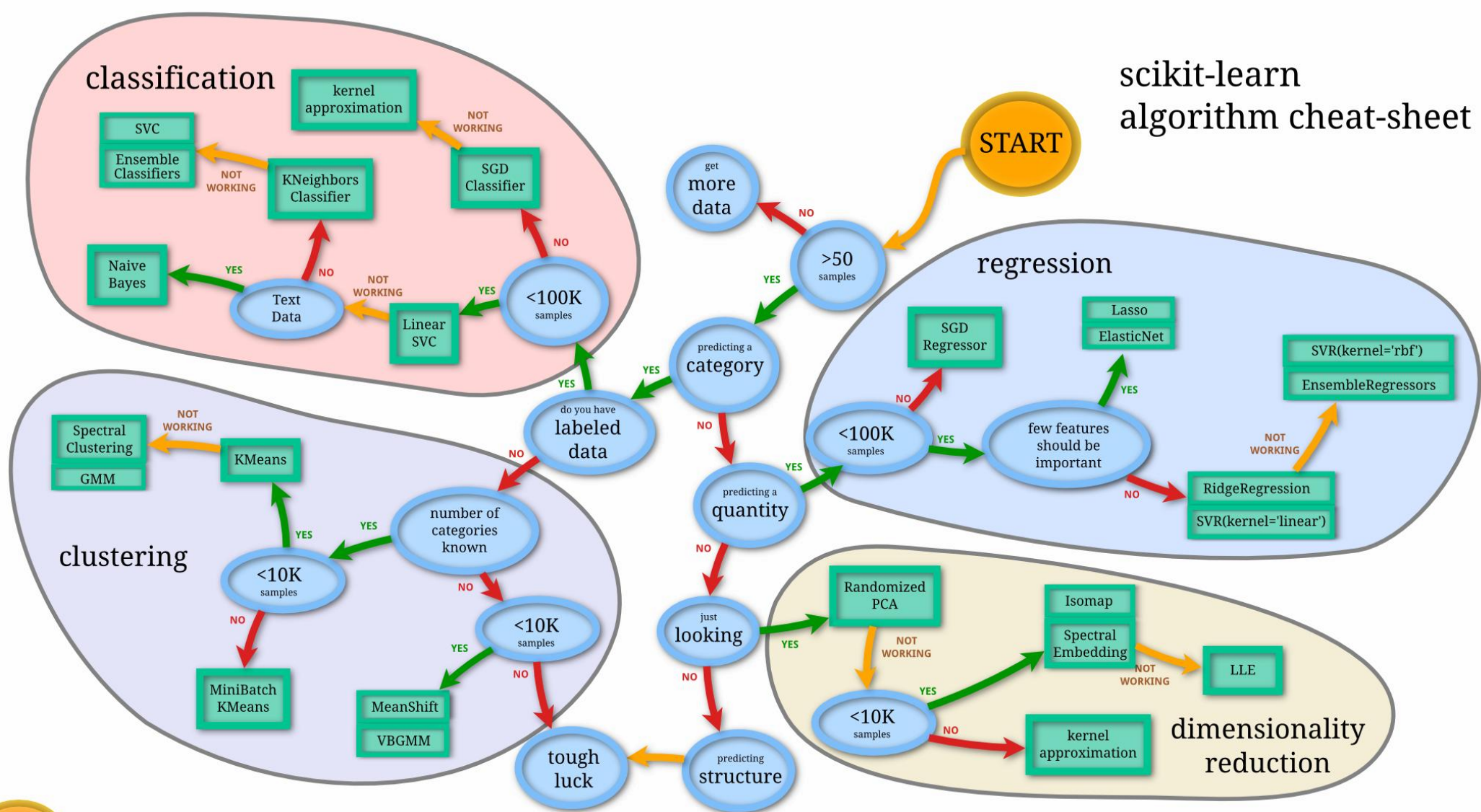


Image from: https://scikit-learn.org/stable/tutorial/machine_learning_map/index.html

ML Textbooks

1. Understanding Machine Learning from Theory to Algorithms 2014. First four chapters useful to read. Remainder chapters are advanced concepts and requires multiple reads. [link](#)
2. All of Statistics 2004. Suitable if you are interested in statistical modelling perspective of ML. Requires multiple reads. [link](#)
3. Elements of statistical learning 2009. <https://web.stanford.edu/~hastie/ElemStatLearn/>
4. Probabilistic Machine Learning: An Introduction 2022. Recent book. <https://probml.github.io/pml-book/book1.html>
5. Practical programming ML books? Package tutorials, online resources, inspecting code on Github.

Review Syllabus

Discuss who should take this course